Intent Mapping for Risk Detection

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🔐 *Data Safety Disclaimer*

*This framework is **fully abstracted**.*

- *No real customer data*
- *No identifiers*
- *No internal thresholds*
- *No proprietary systems*

*It is intended to demonstrate **methodology and thinking**, not expose sensitive information.*

Purpose of the Framework

*This framework provides a **structured method** to convert raw transactional activity into **intent-based risk understanding and controllable detection logic**. It was designed to address a common gap in fraud and AML systems:*

- *Risk often emerges through **sequences of behaviour over time**, not isolated anomalies.*
- *By focusing on **behavioural progression and intent**, this framework enables earlier, more precise, and more explainable detection.*

Core Idea

Instead of analysing transactions one by one, the framework:

1. *Abstracts transactions into **behavioural signals***
2. *Groups signals into **phases***
3. *Infers **intent** from phase-wise behaviour*
4. *Converts intent into **detection hypotheses and rules***

*This ensures every investigation contributes to **system-level learning**, not just case resolution.*

🔍 *Step 1: Behavioural Signal Extraction*

*Raw transactions are converted into **simple behavioural flags** that describe what is happening, not who is involved or how much money moved.*

Examples of Behavioural Flags

- ***Dormancy Break** – Account becomes active after a long inactive period*
- ***Burst Activity** – Multiple transactions in a short window*
- ***FIFO / Self-routing** – Rapid credit → debit cycles*
- ***Unknown Counterparty Spike** – Sudden increase in unfamiliar senders*
- ***Cash-out Proxy** – Rapid withdrawal following inflow*
- ***Device / Environment Change** – Sudden change in access context*
- ***Inactivity Patch** – Deliberate low activity after action*

At this stage:

- *Absolute values are ignored*
- *Identities are abstracted*
- *Only **patterns and transitions** matter*

*The output is a **timeline of flagged events**, not raw transactions.*

🔍 *Step 2: Behaviour Decomposition into Phases*

*Flagged events are grouped into **behavioural phases**. A **phase** represents a time segment where the account shows a **stable pattern of behaviour**.*

Phase Design Principles

- *Combine adjacent events with similar behaviour*
- *Split phases when behaviour meaningfully changes*
- *Merge minor noise into dominant phases*
- *Aim for **3–6 phases per account***

Each phase captures:

- *Dominant behavioural flags*

- *Duration and stability*
- *Whether behaviour is escalating, cooling off, or repeating*

*This converts noisy timelines into a **readable behavioural story**.*

🔍 *Step 3: Intent Mapping*

*Each phase is mapped to a **likely intent**, forming an **Intent Map**.*

Structure

Each phase answers:

- *What was the dominant behaviour?*
- *What was the likely intent?*
- *Was this testing, execution, evasion, or maintenance?*
- *What is the most likely next action?*

An intent map typically includes:

- *Phase boundaries*
- *Behaviour summary*
- *Inferred intent*
- *Risk assessment*
- *Expected next scenario*

*The intent map becomes a **narrative explanation** of how risk evolved over time.*

🔍 *Step 4: Rule & Hypothesis Generation*

*Every completed intent map must produce **at least one detection hypothesis**.*

Rule Design Principles

- *Use **2–3 strong behavioural flags**, not many weak ones*
- *Prefer **relative patterns** over static thresholds*
- *Focus on **controllability** (precision vs false positives)*

- *Describe expected behaviour, not just conditions*

Typical Rule Outputs

- *Objective (what the rule should detect)*
- *Applicable behaviour or persona*
- *Core logic (flag combinations)*
- *Expected detection behaviour*
- *False-positive risk assessment*

This ensures investigations directly improve ***preventive detection systems****.*

🔑 *Key Design Principles*

- *Intent over anomaly*
- *Sequence over snapshot*
- *Behaviour change over absolute values*
- *Explainability over black-box scoring*
- *Human-in-the-loop for high-risk decisions*

Outcomes of Using This Framework

- *Standardise how investigations are analysed*
- *Reduce analyst subjectivity*
- *Convert cases into reusable detection logic*
- *Detect structured misuse earlier*
- *Improve precision while limiting over-blocking*

*Most importantly, it creates a **feedback loop**: Investigations → Intent → Rules → Fewer future cases*
